# Improved Multimodal Emotion Recognition using EEG, ECG, and HRV Signals on the DREAMER Dataset with Attention-Based Deep Fusion

## 1. Introduction

Emotion recognition from physiological signals has emerged as a major research direction in affective computing, human-computer interaction, healthcare monitoring, stress assessment, and brain-computer interfaces. Unlike facial expressions or speech-based approaches, physiological signals provide objective and continuous measurements of internal emotional states and are less susceptible to voluntary manipulation.

This work presents an improved multimodal deep learning framework for emotion recognition using the DREAMER dataset. The proposed architecture integrates three complementary physiological modalities:

- Electroencephalogram (EEG)
- Electrocardiogram (ECG)
- Heart Rate Variability (HRV)

Compared to the previous framework, the proposed system introduces several major architectural improvements:

- Multi-scale EEG temporal feature extraction
- Separate ECG and HRV representation learning
- Attention-guided multimodal fusion
- Deep hierarchical feature compression
- Improved regularization and optimization strategies

The proposed architecture was trained for three-class valence classification and achieved strong performance:

- Accuracy: 91.53%
- Precision: 91.55%
- Recall: 91.53%
- F1-Score: 91.54%

These results demonstrate the effectiveness of attention-based multimodal physiological learning for affective computing applications.

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# 2. DREAMER Dataset Overview

The DREAMER dataset is a multimodal physiological emotion recognition database designed for affective computing research.

The dataset contains recordings from 23 participants exposed to 18 emotional video stimuli designed to induce varying emotional responses.

## 2.1 Physiological Modalities

### EEG Signals

EEG recordings were acquired using the Emotiv EPOC headset with 14 channels:

- AF3
- F7
- F3
- FC5
- T7
- P7
- O1
- O2
- P8
- T8
- FC6
- F4
- F8
- AF4

Sampling frequency:

- 128 Hz

### ECG Signals

ECG recordings were obtained using the Shimmer sensor platform.

Sampling frequency:

- 256 Hz

### Emotional Ratings

After each emotional stimulus, participants evaluated their emotional states using the Self-Assessment Manikin (SAM) protocol.

Three emotional dimensions were recorded:

- Valence
- Arousal
- Dominance

Each score ranged from 1 to 5.

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# 3. Data Preprocessing Pipeline

Proper preprocessing is essential for physiological emotion recognition because EEG and ECG recordings contain noise, motion artifacts, and baseline drift.

The following preprocessing pipeline was implemented.

## 3.1 Bandpass Filtering

### EEG Filtering

A fourth-order Butterworth bandpass filter was applied to EEG signals:

- Low cutoff frequency: 1 Hz
- High cutoff frequency: 45 Hz

This filtering stage removes:

- Baseline drift
- High-frequency noise
- Non-physiological artifacts

while preserving emotion-related EEG frequency components.

### ECG Filtering

ECG signals were filtered using:

- Low cutoff frequency: 0.5 Hz
- High cutoff frequency: 40 Hz

This preserves cardiac morphology while reducing motion artifacts and baseline wander.

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# 3.2 Z-Score Normalization

Each channel was independently normalized using z-score normalization:

$$X_{norm} = \frac{X - \mu}{\sigma}$$

This normalization:

- Stabilizes neural network optimization
- Reduces inter-subject variability
- Prevents scale imbalance between modalities

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# 3.3 Sliding Window Segmentation

Signals were segmented using overlapping temporal windows.

Window configuration:

- Window duration: 10 seconds
- Overlap ratio: 50%

### EEG Window Size

Since EEG was sampled at 128 Hz:

$$128 \times 10 = 1280$$

Each EEG window contained:

- 1280 samples

### ECG and HRV Window Size

Since ECG was sampled at 256 Hz:

$$256 \times 10 = 2560$$

Each ECG and HRV window contained:

- 2560 samples

The overlapping segmentation strategy increased dataset diversity and improved temporal representation learning.

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# 4. HRV Signal Extraction

Heart Rate Variability (HRV) was extracted from ECG recordings to capture autonomic nervous system activity related to emotional regulation.

## 4.1 HRV Extraction Procedure

### Step 1: R-Peak Detection

R-peaks were detected from filtered ECG signals using peak detection algorithms.

### Step 2: RR Interval Computation

RR intervals were computed as:

$$RR_i = t_{i+1} - t_i$$

where $t_i$ represents consecutive R-peak timestamps.

### Step 3: Continuous HRV Interpolation

The RR interval sequence was interpolated into a continuous HRV waveform using linear interpolation.

The resulting HRV signal was used as an independent physiological modality.

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# 5. Label Generation

This work focused on three-class valence classification.

Original valence scores were transformed into categorical emotion labels.

| Original Score | Class |
|---|---|
| 1–2 | Low |
| 3 | Medium |
| 4–5 | High |

This transformation simplifies the classification problem while preserving emotional semantics.

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# 6. Dataset Preparation

Each training sample contained:

- EEG window
- ECG window
- HRV window
- Valence label

The dataset was organized into multimodal tensors for deep learning.

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# 7. Proposed Improved Multimodal Architecture

The proposed architecture introduces an attention-based multimodal deep learning framework composed of:

1. Multi-scale EEG pathway
2. ECG pathway
3. HRV pathway
4. Attention-based fusion module
5. Deep classification head

The overall architecture jointly learns neural and cardiac emotional representations.

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# 8. Multi-Scale EEG Pathway

The EEG pathway was designed to capture emotional patterns at multiple temporal resolutions.

Instead of using a single CNN stream, the architecture employs three parallel CNN branches with different receptive fields:

- Small-scale branch
- Medium-scale branch
- Large-scale branch

This design enables simultaneous learning of:

- Short-term temporal patterns
- Mid-range dependencies
- Long-range EEG dynamics

## 8.1 EEG Branch Architecture

Each branch contains:

### Convolution Block 1

- Conv1D
- Batch Normalization
- ReLU
- MaxPooling

### Convolution Block 2

- Conv1D
- Batch Normalization
- ReLU
- MaxPooling

### Convolution Block 3

- Conv1D
- Batch Normalization
- ReLU

### Global Feature Aggregation

Adaptive average pooling was applied to generate compact feature representations.

Each branch outputs a 512-dimensional feature vector.

The outputs from all three branches are concatenated and compressed into a unified 512-dimensional EEG embedding.

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# 9. ECG Pathway

The ECG pathway processes raw cardiac electrical activity signals.

The architecture learns emotion-related cardiac morphology and rhythm variations directly from raw ECG waveforms.

## 9.1 ECG CNN Architecture

The ECG branch contains three convolutional blocks:

### Block 1

- Conv1D: 1 → 64
- Kernel size: 15
- Stride: 2

### Block 2

- Conv1D: 64 → 128
- Kernel size: 7
- Stride: 2

### Block 3

- Conv1D: 128 → 256
- Kernel size: 3

Global average pooling is used before feature projection.

The ECG branch outputs a 512-dimensional representation.

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# 10. HRV Pathway

The HRV pathway models autonomic nervous system variability associated with emotional responses.

## 10.1 HRV CNN Architecture

The HRV branch consists of:

### Block 1

- Conv1D: 1 → 32
- Kernel size: 11

### Block 2

- Conv1D: 32 → 64
- Kernel size: 7

### Block 3

- Conv1D: 64 → 128
- Kernel size: 5

After global pooling and feature projection, the HRV branch outputs a 256-dimensional embedding.

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# 11. ECG-HRV Fusion Pathway

The ECG and HRV feature vectors are concatenated:

$$512 + 256 = 768$$

A compression layer projects the fused cardiac representation into a shared 512-dimensional embedding.

This enables joint modeling of:

- Cardiac electrical activity
- Autonomic variability dynamics

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# 12. Attention-Based Multimodal Fusion

A lightweight self-attention fusion mechanism was introduced to adaptively combine EEG and ECG-HRV representations.

Unlike conventional concatenation-based fusion, the proposed mechanism learns modality importance dynamically.

## 12.1 Self-Attention Gate

Each modality passes through a self-attention gating module consisting of:

- Query projection
- Key projection
- Value projection

Attention scores are computed using similarity between query and key representations.

The generated attention weights scale modality-specific features.

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# 12.2 Adaptive Fusion Weights

Learnable multilayer perceptrons estimate modality importance scores:

- EEG importance
- ECG-HRV importance

Softmax normalization produces adaptive fusion coefficients:

$$\alpha_{EEG}, \alpha_{ECG}$$

The final fused representation is computed as:

$$F = \alpha_{EEG} A_{EEG} + \alpha_{ECG} A_{ECG}$$

This mechanism allows the network to dynamically emphasize the most informative modality.

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# 13. Classification Head

The fused representation is processed using a deep fully connected classifier.

## 13.1 Architecture

### Fully Connected Layer 1

- 512 → 256
- Batch Normalization
- ReLU
- Dropout: 0.3

### Fully Connected Layer 2

- 256 → 128
- Batch Normalization
- ReLU
- Dropout: 0.3

### Output Layer

- 128 → 3 classes

The output layer predicts:

- Low valence
- Medium valence
- High valence

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# 14. Training Strategy

## 14.1 Dataset Split

The dataset was divided using stratified sampling:

| Split | Percentage |
|---|---|
| Training | 70% |
| Validation | 15% |
| Testing | 15% |

Stratification preserved class balance across all subsets.

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# 14.2 Hyperparameters

| Parameter | Value |
|---|---|
| Batch Size | 32 |
| Maximum Epochs | 100 |
| Learning Rate | $1 \times 10^{-3}$ |
| Minimum Learning Rate | $1 \times 10^{-6}$ |
| Weight Decay | $1 \times 10^{-4}$ |
| Dropout Rate | 0.3 |
| Label Smoothing | 0.1 |
| Early Stopping Patience | 10 |

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# 14.3 Optimization

The model was trained using the Adam optimizer.

A ReduceLROnPlateau scheduler was applied to reduce the learning rate when validation performance stagnated.

Regularization techniques included:

- Batch normalization
- Dropout
- Weight decay
- Label smoothing
- Early stopping

These strategies improved model generalization and reduced overfitting.

# 15. Evaluation Metrics

The following metrics were used:

- Accuracy
- Precision
- Recall
- F1-score
- Classification report

Weighted averaging was used for multi-class evaluation.

# 16. Experimental Results

The proposed attention-based multimodal framework achieved strong performance on the DREAMER dataset.

## 16.1 Overall Performance

| Metric | Score |
|---|---|
| Accuracy | 0.9153 |
| Precision | 0.9155 |
| Recall | 0.9153 |
| F1-Score | 0.9154 |

# 16.2 Detailed Classification Report

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Low | 0.9165 | 0.9231 | 0.9198 | 975 |
| Medium | 0.8842 | 0.9004 | 0.8922 | 492 |
| High | 0.9296 | 0.9150 | 0.9222 | 1024 |

## 16.3 Overall Statistics

| Metric | Value |
|---|---|
| Accuracy | 0.9153 |
| Macro Average F1 | 0.9114 |
| Weighted Average F1 | 0.9154 |

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# 17. Discussion

The obtained results demonstrate that attention-guided multimodal fusion significantly improves emotion recognition performance.

Several key observations can be identified.

## 17.1 Effectiveness of Multi-Scale EEG Learning

The multi-branch EEG architecture successfully captured emotional dynamics across multiple temporal resolutions.

This improved representation learning compared to single-scale CNN processing.

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## 17.2 Complementarity of ECG and HRV

ECG and HRV signals provided complementary cardiac and autonomic information.

The dedicated cardiac fusion pathway enabled the model to jointly learn:

- Heartbeat morphology
- Rhythm dynamics
- Autonomic variability

which improved emotional discrimination.

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## 17.3 Benefits of Attention-Based Fusion

The attention fusion mechanism dynamically weighted modality importance during inference.

This adaptive fusion strategy improved robustness and representation quality compared to static concatenation methods.

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## 17.4 Strong Overall Classification Performance

The proposed framework achieved:

- 91.53% accuracy
- 91.54% weighted F1-score

representing a substantial improvement over the previous baseline architecture.

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# 18. Conclusion

This work presented an improved multimodal deep learning framework for emotion recognition using EEG, ECG, and HRV signals from the DREAMER dataset.

The proposed system introduced several architectural improvements:

- Multi-scale EEG representation learning
- Dedicated ECG-HRV cardiac modeling
- Attention-guided multimodal fusion
- Deep hierarchical classification

A comprehensive preprocessing pipeline was implemented including:

- Signal filtering
- HRV extraction
- Temporal segmentation
- Signal normalization

The proposed model achieved:

- 91.53% classification accuracy
- 91.54% weighted F1-score

The results demonstrate that attention-based multimodal physiological learning provides highly discriminative emotional representations and significantly improves affective computing performance.

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# 19. Future Work

Several directions may further improve the framework:

- Subject-independent cross-validation
- Transformer-based multimodal fusion
- Temporal sequence modeling using LSTM or GRU networks
- Graph neural networks for EEG connectivity modeling
- Cross-attention multimodal transformers
- Real-time emotion recognition systems

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# 20. Technologies and Libraries

The framework was implemented using:

- Python
- PyTorch
- NumPy
- SciPy
- Scikit-learn
- Matplotlib
- Seaborn

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# 21. Summary

This work demonstrates that combining EEG, ECG, and HRV signals using an attention-based multimodal deep learning framework can achieve highly accurate emotion recognition.

The proposed architecture successfully learned complementary neural and cardiac emotional representations and achieved state-of-the-art performance improvements over the previous baseline model.

The framework provides a strong foundation for future multimodal affective computing and physiological artificial intelligence research.